

Name:

Form:.....

**ASCHAM SCHOOL
MATHEMATICS EXAMINATION
FORM 6 - 3 UNIT
1999**

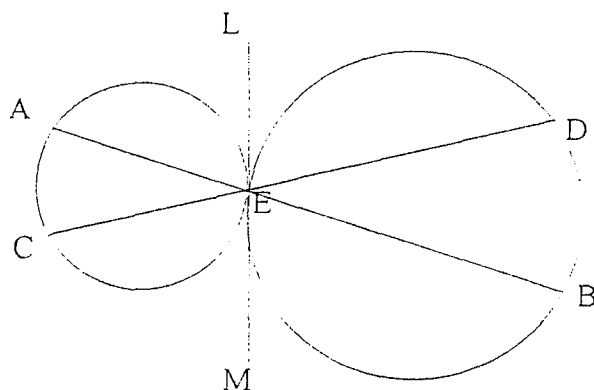
July 1999

Time allowed: 2 hours

- * All questions should be attempted
- * All necessary working must be shown
- * All questions are of equal value
- * Marks may not be awarded for careless or badly arranged work.
- * Write your name on each booklet clearly marked:
Question 1, Question 2, etc.
- * Begin each question in a new booklet.
- * Approved calculators may be used.
- * Copies of diagrams for all questions are provided on pages
11-14 in order to save time. You may use them but **you
must staple them into your booklets.**

Question 1 Marks:

- (a) Find the acute angle, to the nearest degree, between the lines $y = 3x + 1$ and $y = -x + 6$ 3
- (b) Solve the inequality $\frac{1}{x+1} < 3, \quad x \neq -1$ 3
- (c) Find the coordinates of the point P which divides the interval AB with end points A(-1, 2) and B(3, -5) internally in the ratio 2:3. 2
- (d) Use the substitution $u = t + 1$ to evaluate $\int_0^1 \frac{t}{\sqrt{t+1}} dt$ 3
- (e) Two circles touch externally at E. 3



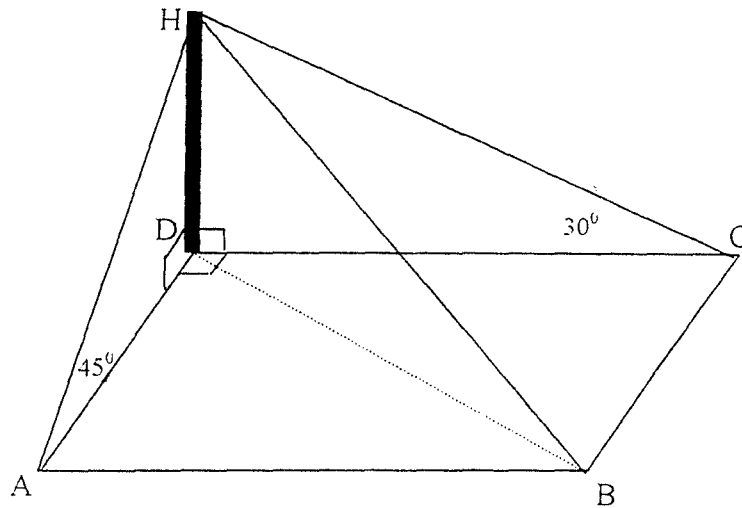
(A copy of the diagram above is on page 10.)

AB and CD intersect at E. LM is a common tangent at E

Prove that AC is parallel to DB.

Question 2

Marks:



- (a) A post HD stands vertically at one corner of a rectangular field $ABCD$. The angles of elevation of the top H of the post from the nearest corners A and C respectively are 30° and 45° .
(A copy of the diagram above is on page 13.)
- (i) If $AD = a$ units, find the length of BD in terms of a . 2
- (ii) Hence find the angle of elevation of H from the corner B to the nearest minute. 1
- (b) Taking $x = -\frac{\pi}{6}$ as a first approximation to the root of the equation $2x + \cos x = 0$, use Newton's method once to show that a better approximation to the root of the equation is $\frac{-\pi - 6\sqrt{3}}{30}$ 4
- (c) (i) Find the domain and range of $f^{-1}(x) = \sin^{-1}(3x - 1)$ 2
- (ii) Sketch the graph of $y = f^{-1}(x)$. 2
- (iii) Find the equation representing the inverse function $f(x)$ and state the domain and range. 3

Question 3

Marks:

- (a) (i) Express $3\sin x - \sqrt{3}\cos x$ in the form $A\sin(x - \alpha)$, where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 3
- (ii) Determine the minimum value of $3\sin x - \sqrt{3}\cos x$. 1
- (iii) Solve $3\sin x - \sqrt{3}\cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. 3

- (b) Newton's Law of cooling states that the rate of cooling of a body is proportional to the excess of the temperature of a body above the surrounding temperature. This rate can be expressed by the differential equation:

$$\frac{dT}{dt} = -k(T - T_0),$$

where T is the temperature of the body, T_0 is the temperature of the surroundings, t is the time in minutes and k is a constant.

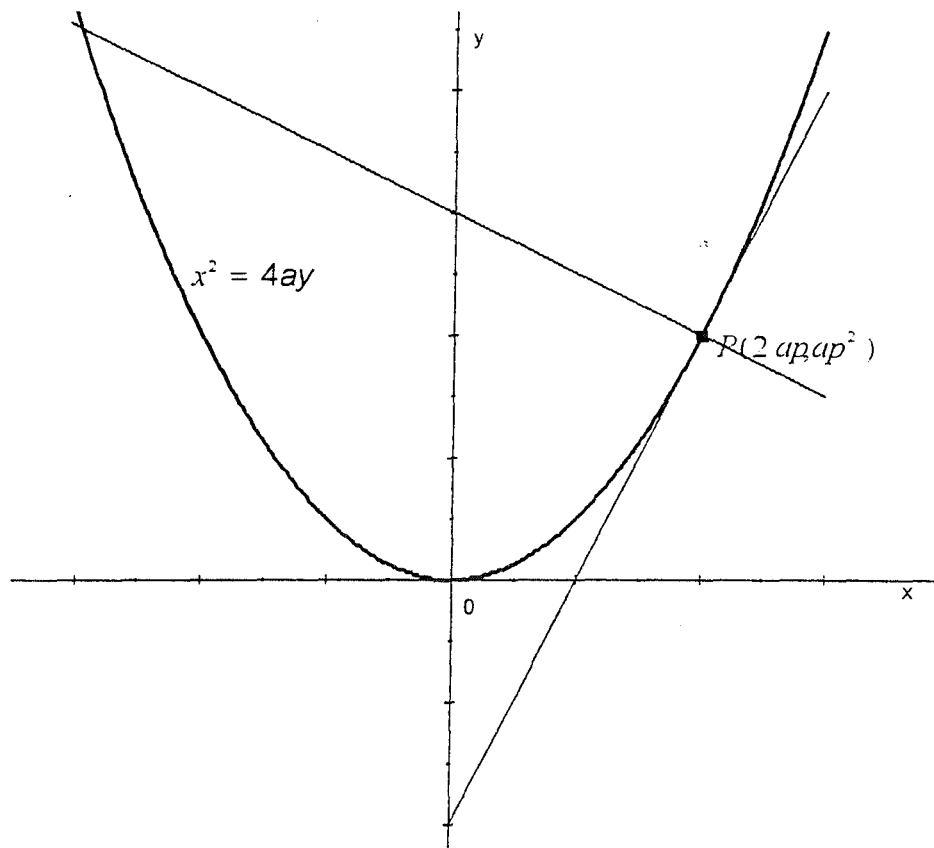
- (i) Show that $T = T_0 + Ae^{-kt}$, where A is a constant, is a solution of the differential equation $\frac{dT}{dt} = -k(T - T_0)$. 2

- (ii) A cup of tea cools from 85°C to 80°C in 1 minute at a room 5

temperature of 25°C . Find the temperature of the cup of tea after a further 4 minutes have elapsed. Answer to the nearest degree.

Question 4

Marks:



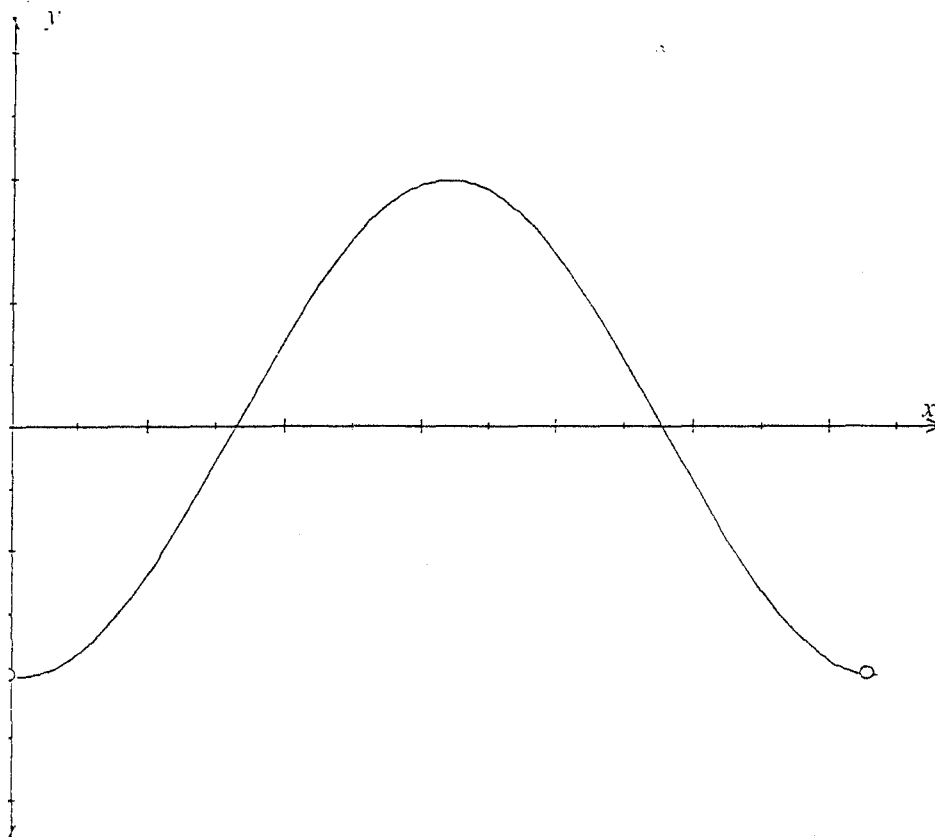
- (a) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. 3
Show the equation of the normal to the parabola at P is $x + py = 2ap + ap^3$.
- (b) Write down the equation of the normal to the parabola at Q . The normals intersect at N . Find the coordinates of N . 3
- (c) Show the equation of the chord PQ is $y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$ 3
and determine the condition necessary for PQ to be a focal chord.
- (d) If PQ is a focal chord and N is the intersection of the normals, find the equation of the locus of N . 4
- (e) (A copy of the diagram above is on page 11.) 1
On the diagram above, the tangent and normal are drawn at P .
Mark clearly on your own diagram the points Q and N which correspond to P .

Question 5

Marks:

- (a) The graph of $x = -a \cos nt$ for $0 \leq t \leq \frac{2\pi}{n}$ is drawn below. (A copy of the diagram above is on page 12.) Label axes and show intercepts accurately.

2



- (b) On a certain day the depth of water in a harbour at low tide at 4:30 am is 5 metres. At the following high tide at 10:45 am the depth is 15 metres. Assuming the rise and fall of the surface of the water to be simple harmonic, find between what times during the morning a ship may safely enter the harbour if the minimum depth of $12\frac{1}{2}$ metres of water is required.

6

- (c) Given that $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(2-x)$ have values for $0 \leq x \leq \frac{\pi}{2}$

(i) show that $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$

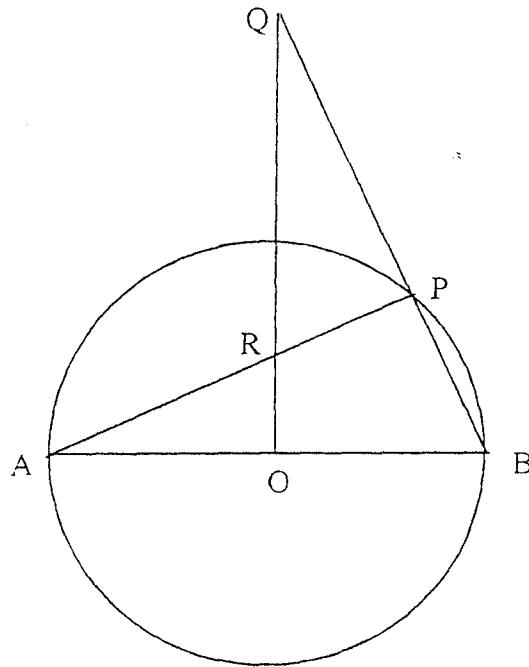
3

(ii) Hence, or otherwise, solve the equation $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(2-x)$

3

Question 6**Marks:**

- (a) O is the centre of the circle. BPQ is a straight line ORQ is perpendicular to AOB as shown below.



(A copy of the diagram above is on page 14.)

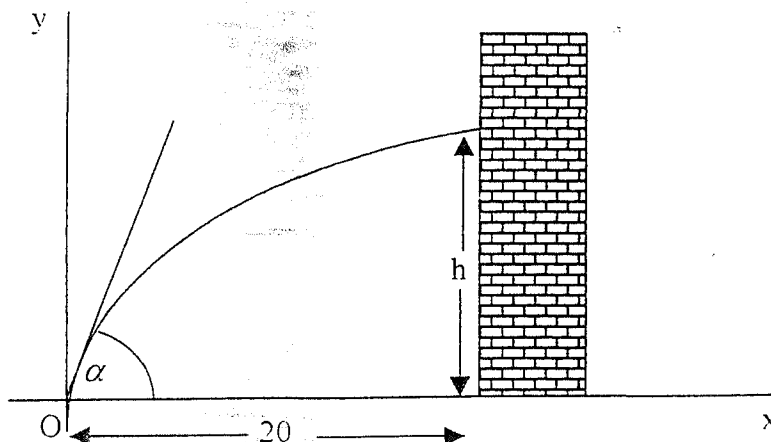
Prove that:

- | | |
|-------------------------------------|---|
| (i) A, O, P, Q are concyclic, and | 3 |
| (ii) $\angle OPA = \angle OQB$. | 2 |
- (b) Prove by using mathematical induction that $5^n \geq 1 + 4n$, for $n > 1$, $n \in J^+$. 4
- (c) The cubic equation $2x^3 - x^2 + x - 1 = 0$ has roots α , β , and γ . Evaluate
- | | |
|---|---|
| (i) $\alpha\beta + \beta\gamma + \alpha\gamma$ | 1 |
| (ii) $\alpha\beta\gamma$ | 1 |
| (iii) $\alpha^2\beta^2\gamma + \beta^2\gamma^2\alpha + \alpha^2\gamma^2\beta$ | 1 |
- (d) The equation $2\cos^3\theta - \cos^2\theta + \cos\theta - 1 = 0$ has roots $\cos a$, $\cos b$ and $\cos c$. 2
- Using appropriate information from (c) above prove that
- $$\sec a + \sec b + \sec c = 1$$

Question 7**Marks:**

A softball player hits the ball from ground level with a speed of 20 ms^{-1} and an angle of elevation α . It flies toward a high wall 20 m away on level ground.

- (a) Taking the origin at the point where the ball is hit, derive expressions for 3



the horizontal and vertical components x and y of displacement at time t seconds. Take $g = 10 \text{ ms}^{-2}$.

- (b) Hence find the equation of the path of the ball in flight in terms of 1
 x , y and α .
- (c) Show that the height h at which the ball hits the wall is given by 2
$$h = 20 \tan \alpha - 5(1 + \tan^2 \alpha).$$
- (d) Using part (c) above, show that the maximum value of h occurs 2
when $\tan \alpha = 2$.
- (e) Find 6
(i) this maximum height h ,
(ii) the speed and the angle at which the ball hits the wall in this case.

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$$y = 3x + 2 \quad m_1 = 3 \quad y = 1 - x \quad m_2 = -1$$

$$\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{3 + 1}{1 + 3(-1)} \right|$$

$$= 2 \checkmark$$

$$\therefore \alpha = 63^\circ 26'$$

$$= 63^\circ \text{ (nearest deg.)}$$

$$\frac{1}{x+1} < 3 \Rightarrow x+1 < 3(x+1)^2$$

$$= 3(x^2 + 2x + 1)$$

$$3x^2 + 5x + 2 > 0$$

$$(3x+2)(x+1) > 0$$

$$x < -\frac{1}{2} \text{ or } x > -\frac{2}{3}$$

2) 3, -5

$$P\left(\frac{2 \times 3 - 3 \times 1}{5}, \frac{2 \times -5 + 3 \times 2}{5}\right) \checkmark$$

$$P\left(\frac{3}{5}, -\frac{4}{5}\right) \checkmark$$

$$\int_0^1 \frac{1}{\sqrt{t+1}} dt$$

$$= \int_1^2 \frac{u-1}{\sqrt{u}} du \checkmark$$

$$u = t+1 \quad \text{if } t=0 \quad u=1$$

$$t = u-1 \quad \text{if } t=1 \quad u=2$$

$$du = dt$$

$$= \int_1^2 (u^{\frac{1}{2}} - u^{-\frac{1}{2}}) du$$

$$= \left[\frac{2u^{\frac{3}{2}}}{3} - 2u^{\frac{1}{2}} \right]_1^2 \checkmark$$

$$= \left(\frac{4\sqrt{2}}{3} - 2\sqrt{2} \right) - \left(\frac{2}{3} - 2 \right)$$

$$= -\frac{2\sqrt{2}}{3} + \frac{4}{3} \checkmark$$

$$= \frac{4-2\sqrt{2}}{3} \text{ (OR)}$$

Q1. (e) $\angle AEL = \angle ALE$ ($\angle$ in alt. segm.) $\checkmark$
 $\angle AEL = \angle MEB$ (vert. opp. $\angle$ s) $\checkmark$
 $\angle BDE = \angle MEB$ ($\angle$ in alt. segm.) $\checkmark$
 $\therefore \angle ACE = \angle BDE$ $\checkmark$

But $\angle ACE$ is alternate to $\angle BDE \therefore AC \parallel DB \checkmark$

Q2. (a) In $\triangle ADH$ $AD = DH = a$
 In $\triangle HDC$ $\tan 30^\circ = \frac{a}{x}$

$$\therefore DC = a\sqrt{3} \checkmark$$

In $\triangle BDC$ $BD^2 = DC^2 + CB^2$
 $= (a\sqrt{3})^2 + a^2$
 $= 4a^2$

$$\therefore BD = 2a \checkmark$$

(i) In $\triangle HDB$ $\frac{HD}{BD} = \tan \hat{HBD}$
 $\frac{a}{2a} = \tan \hat{HBD}$
 $\therefore \angle HBD = 26^\circ 34' \text{ (to nearest minute)}$

(p) $\frac{d}{dx} (2x + \cos x) = 2 - \sin x \checkmark$

$$f\left(\frac{\pi}{6}\right) = 2 + \sin \frac{\pi}{6}$$

$$= 2 + \frac{1}{2}$$

$$= \frac{5}{2} \checkmark$$

$$f\left(\frac{\pi}{3}\right) = \frac{5}{2} - \frac{\sqrt{3}}{2}$$

$$= \frac{5\sqrt{3}-2\sqrt{3}}{2} \checkmark$$

$$Z_1 = -\frac{\sqrt{3}}{6} - \frac{3\sqrt{3}-2\sqrt{3}}{6} \times \frac{2}{5} \checkmark$$

$$= -\frac{\sqrt{3}}{6} - \frac{2\sqrt{3}}{15} + \frac{2\sqrt{3}}{15}$$

$$= -\frac{\sqrt{3}}{30} \checkmark$$

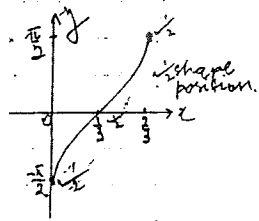
$$y = \sin^{-1}(3x-1)$$

$$\text{Domain of } f^{-1}(x): -1 \leq 3x-1 \leq 1$$

$$0 \leq 3x \leq 2$$

$$0 \leq x \leq \frac{2}{3}$$

$$\text{Range of } f^{-1}(x): -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



$y = \sin^{-1}(3x-1)$ is the inverse function of $f(x)$

$\sin y = 3x-1$ is the inverse of $f(x)$

$$\sin x = 3y-1$$

$$y = \frac{1}{3} + \frac{1}{3} \sin x \text{ is } f(x)$$

$$\text{or } f(x) = \frac{1}{3} + \frac{1}{3} \sin x. \text{ Its domain } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \text{ and range } 0 \leq y \leq \frac{2}{3}$$

$$3 \sin x - \sqrt{3} \cos x = A \sin(x-\alpha)$$

$$= A \sin x \cos \alpha - A \cos x \sin \alpha$$

$$\frac{A \sin \alpha}{A \cos \alpha} = \frac{\sqrt{3}}{3} \therefore \tan \alpha = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}} \therefore \alpha = \frac{\pi}{6}$$

$$A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = 9 + 3 \therefore A^2 = 12, A = 2\sqrt{3}$$

$$\therefore 3 \sin x - \sqrt{3} \cos x = 2\sqrt{3} \sin(x - \frac{\pi}{6})$$

$$\text{min. value} = -2\sqrt{3}$$

$$2\sqrt{3} \sin(x - \frac{\pi}{6}) = \sqrt{3}$$

$$\sin(x - \frac{\pi}{6}) = \frac{1}{2}$$

$$x - \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$\therefore x = \frac{\pi}{3} \text{ or } x = \pi \text{ for } 0 \leq x < \pi$$

$$Q3 (b) (i) T = T_0 + Ae^{-kt} \text{ i.e. } T - T_0 = Ae^{-kt}$$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T - T_0)$$

$$(ii) 85 = 25 + Ae^0 \therefore A = 60$$

$$T = 25 + 60e^{-kt}$$

$$80 = 25 + 60e^{-kt}$$

$$e^{-kt} = \frac{55}{60}$$

$$-kt = \ln \frac{11}{12} \text{ (or } kt = \ln \frac{12}{11})$$

$$\therefore T = 25 + 60e^{(\ln \frac{12}{11})t}$$

$$\text{when } t=5 \quad T = 25 + 60e^{-5 \ln \frac{12}{11}}$$

$$\approx 64$$

$T \approx 64^\circ$ after a further 4 minutes.

$$Q.4 (a) x^2 = 4ay \quad \frac{dy}{dx} = \frac{x}{2a} \text{ at } x = 2ap$$

$$\frac{dy}{dx} = p$$

$$\therefore \text{gradient of normal} = -\frac{1}{p}$$

$$\text{Eqn: } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

Normal at P $x+py = 2ap + ap^3$ (1)
 " Q $x+qy = 2aq + aq^3$ (2)

(2) $y(p-q) = 2a(p-q) + a(p^3 - q^3)$
 $a = 2a + a(p^2 + q^2 + pq)$
 $y = a(p^2 + q^2 + pq + 2)$ ✓

$q \cdot xq + py = 2apq + ap^3$
 $p \cdot xp - pqy = -2apq - apq^3$

$x(q-p) = apq(p-q)(p+q)$

$x = -apq(p+q)$ ✓

$[-apq(p+q), a(p^2 + q^2 + pq + 2)] = N(X, Y)$

$m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq} \Rightarrow m_{PQ} = \frac{x(p+q)(p-q)}{2a(p-q)}$
 $= \frac{p+q}{2} \cdot 1$ ✓

hold PQ: $y - ap^2 = \frac{p+q}{2}(x - 2ap)$ ✓

cf. S(0, a): $a - ap^2 = \frac{p+q}{2}(-2ap)$

$a - ap^2 = -ap^2 - apq$

$\therefore pq = -1$ if PQ is through S. ✓

$X = -apq(p+q) \Rightarrow X = -a(p+q) \Rightarrow p+q = \frac{X}{a}$ (1)

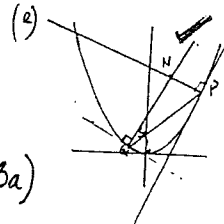
$Y = a(p^2 + q^2 + pq + 2) \Rightarrow Y = a(p^2 + q^2 + 1) \Rightarrow p^2 + q^2 = \frac{Y}{a} - 1$ (2)

$p^2 + q^2 = (p+q)^2 - 2$ ✓

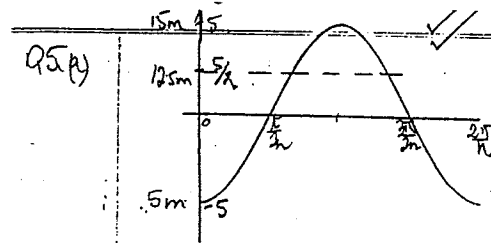
(3) $\frac{Y}{a} - 1 = \frac{X^2}{a^2} + 2$

$\frac{Y}{a} = \frac{X^2}{a^2} + 3$

✓ $Y = \frac{X^2}{a} + 3a$ or $X^2 = a(Y - 3a)$



(3)



$\therefore x = -5 \cos \frac{4\pi}{25} t$ ✓

when $x = \frac{5}{2}$

$\frac{5}{2} = -5 \cos \frac{4\pi}{25} t$

$-\frac{1}{2} = \cos \frac{4\pi}{25} t$ ✓

$\frac{4\pi}{25} t = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \dots$

$t = \frac{25}{6}, \frac{25}{3}, \dots$ ✓

$\therefore$ the times between which the ship may enter the harbour are 8:40 am and 12:50 pm. ✓

(4) (i) LHS = $\sin(\sin^{-1}x - \cos^{-1}x)$

$= \sin(\alpha - \beta)$

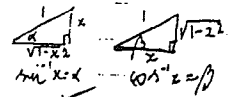
$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$ ✓

$= x \cdot x - \sqrt{1-x^2} \cdot \sqrt{1-x^2}$

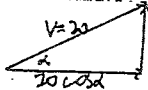
$= 2x^2 - 1$ ✓

$= RHS$

where



(5)

a)  at $t=0$

$$\begin{aligned} x &= 0 & y &= -10 \\ \dot{x} &= 20 \cos \alpha & \dot{y} &= -10t + 20 \sin \alpha \\ x &= 20t \cos \alpha & y &= -5t^2 + 20t \sin \alpha \end{aligned}$$

① ②

Sub $t = \frac{x}{20 \cos \alpha} \rightarrow$ ②

$$y = \frac{x \cdot 20 \sin \alpha}{20 \cos \alpha} - 5 \frac{x^2}{100 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{x^2}{80} \sec^2 \alpha$$

when $x=20$, $y=h$

$$h = 20 \tan \alpha - \frac{400}{80} \sec^2 \alpha$$

$$h = 20 \tan \alpha - 5 \sec^2 \alpha$$

$$\frac{dh}{d\alpha} = 20 \sec^2 \alpha - 10 \sec^2 \alpha \tan \alpha$$

$$10 \sec^2 \alpha (2 - \tan \alpha) = 0 \text{ for max}$$

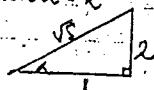
$$\sec \alpha = 0 \text{ or } \tan \alpha = 2$$

$$\alpha = 63^\circ (\text{nearest deg})$$

$$60^\circ \quad 63^\circ \quad 70^\circ$$

$\begin{matrix} 60^\circ & 63^\circ & 70^\circ \\ + & 0 & - \end{matrix}$ $\therefore h$ is max. when $\tan \alpha = 2$

$$h_{\max} = 20 \cdot 2 - 5 \cdot \sqrt{5}^2 = 15 \text{ metres}$$



u.i.v

$$v = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

(16)

$$= \sqrt{(20 \cos \alpha)^2 + (20 \sin \alpha - 10t)^2}$$

To find t :

$$= \sqrt{\left(\frac{20}{\sqrt{5}}\right)^2 + \left(\frac{20}{\sqrt{5}} - 10 \times \frac{1}{\sqrt{5}}\right)^2}$$

$$20 = v \cos \alpha$$

$$20 = 20 \times \frac{1}{\sqrt{5}}$$

$$t = \frac{1}{\sqrt{5}}$$

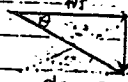
$$= \sqrt{80 + (8\sqrt{5} - 10\sqrt{5})^2}$$

$$= \sqrt{80 + 20}$$

$= 10 \text{ m/s}$ is the speed of ball when hits the wall on its way down.

$$\tan \theta = \frac{2\sqrt{5}}{4\sqrt{5}}$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\right)$$



hits wall at

$$\alpha = \tan^{-1}\left(\frac{15}{20}\right)$$

$$= \tan^{-1} \frac{3}{4}$$

